

Watermark Detection and Removal Using Edge Detection and Adaptive Thresholding

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Abstract—Watermarks are frequently used in digital media for copyright protection. However, watermark removal with image quality maintenance is required in most applications like content reuse and image restoration. A watermark detection and removal approach using edge detection and adaptive thresholding is presented in this paper. Two processes are suggested: one using Canny edge detection with morphological filtering and another using adaptive thresholding with inpainting. Experimental research demonstrates the effectiveness of these methods with different forms of watermarks. The challenges, limitations, and possible future work improvements are considered.

Index Terms—Watermark Removal, Edge Detection, Adaptive Thresholding, Image Inpainting, Morphological Operations

I. INTRODUCTION

Watermarks are usually used to stop digital content from being illegally used. Even though they are very important for copyright enforcement, there are legitimate cases where watermarks must be removed, such as restoration of archives, reuse of content, and enhancing visual attractiveness. The main problem is watermark removal without degrading the original image data. Traditional techniques struggle with watermark types, transparency levels, and complex backgrounds. This project investigates two main methods: edge detection-based segmentation and adaptive thresholding-based segmentation, then inpainting to recover the image.

II. BACKGROUND

Several studies have explored watermark detection and removal techniques:

- Xu et al. (2017) proposed a statistical thresholding approach combined with exemplar-based inpainting to remove watermarks efficiently [1].
- Goh et al. (2018) analyzed the performance of Otsu's thresholding for image segmentation, highlighting its sensitivity to contrast and noise [2].
- AlSaeed et al. (2016) introduced an optimized Otsu method to enhance segmentation efficiency while reducing computational complexity [3].
- Nayak et al. (2015) improved Canny edge detection using morphological filters, which refined detected edges and enhanced object boundary clarity [4].
- Bhatt and Shah (2012) studied various image interpolation techniques to enhance image clarity for improved segmentation [5].

Existing approaches face challenges in handling semi-transparent watermarks and complex backgrounds. Many methods assume structured watermarks with uniform placement, which is not always the case. While edge detection techniques perform well for structured watermarks, they fail when dealing with blended pattern watermarks. Adaptive thresholding provides improved segmentation, but its effectiveness varies depending on image conditions. This project aims to compare and integrate multiple approaches, evaluating their performance across different watermark styles.

III. METHODOLOGY

This project explores two primary watermark detection and removal pipelines, each designed to handle specific types of watermarks commonly encountered in digital media. Structured watermarks, such as bold text or logos, exhibit clear contours and contrast strongly with the image background. In contrast, blended pattern or semi-transparent watermarks are often diffused into the image and require more nuanced segmentation techniques. The system's design accounts for both cases using distinct approaches tailored to maximize segmentation precision and restoration quality.

A. Overview of Approaches

The edge detection-based pipeline is optimized for extracting structured watermark edges. It uses a series of operations including Canny edge detection, morphological enhancement, and contour filtering to isolate the watermark. Once identified, the watermark region is removed using inpainting. This approach is particularly effective for sharp, high-contrast watermarks that produce strong gradient responses.

The adaptive thresholding-based pipeline, on the other hand, is engineered for watermark patterns that lack clear boundaries. It uses a thresholding technique that dynamically adjusts to local image characteristics, followed by morphological processing to clean and consolidate the segmented region. A sharpening filter is then applied to restore clarity in areas affected by inpainting. This pipeline excels in scenarios where the watermark blends with textured or low-contrast backgrounds.

- 1) **Grayscale Conversion:** Grayscale conversion simplifies the input image by collapsing three RGB color channels

into a single luminance channel. This not only reduces computational load but also emphasizes intensity variations useful for detecting watermark features. The standard conversion formula is:

$$I_{gray} = 0.299R + 0.587G + 0.114B$$

- 2) **Canny Edge Detection:** This multi-stage algorithm detects edge-like structures by first applying Gaussian blurring to reduce noise, then computing intensity gradients. It uses non-maximum suppression and double thresholding to identify strong and weak edges, producing a binary edge map that is ideal for structured watermark detection.
- 3) **Morphological Operations:** Morphological filters, such as dilation and closing, are used to refine binary masks. Dilation helps bridge fragmented edges, while closing (a dilation followed by erosion) fills small holes in the mask. These operations are essential for ensuring that the segmented watermark region forms a continuous and well-defined mask.
- 4) **Adaptive Thresholding:** Unlike global thresholding, adaptive methods calculate a threshold value for each pixel based on the intensity values of its neighbors. This enables effective segmentation in regions with varying lighting or contrast. The Gaussian method used here is robust for soft watermark detection.
- 5) **Image Inpainting:** Inpainting refers to the reconstruction of missing or corrupted image regions. This work uses OpenCV's implementation of the Telea inpainting method, which fills the target area by propagating nearby pixel values in an edge-aware manner. This is crucial for removing the watermark while preserving image continuity.
- 6) **Sharpening Filter:** To compensate for the blurring effect caused by inpainting, a sharpening filter is applied. This filter emphasizes high-frequency components, thereby enhancing edge details. The kernel used is:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

B. Pipeline 1: Edge Detection-Based Approach

This pipeline is designed for images with watermarks that have a high degree of structure and contrast. The following operations are performed sequentially:

- 1) The image is converted to grayscale, enhancing edge visibility while simplifying pixel values.
- 2) Canny edge detection is applied to extract the strongest gradients corresponding to watermark outlines.
- 3) Morphological dilation connects fragmented edges, producing a cleaner edge map.
- 4) Contour detection is used to isolate large, contiguous regions. Small contours below a certain threshold are discarded to reduce noise.

- 5) The resulting binary mask is then passed to OpenCV's inpainting function, which reconstructs the masked region using the Telea algorithm.

This approach is highly effective for logos, watermarks with strong geometric shapes, and bold text overlays.

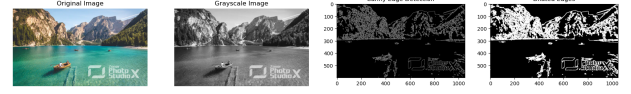


Fig. 1. Left: Original and Grayscale Image. Right: Canny Edge Detection with Dilation

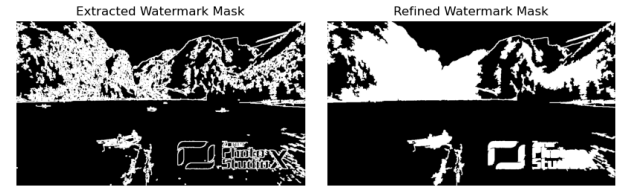


Fig. 2. Left: Initial Watermark Mask. Right: Refined Watermark Mask



Fig. 3. Final Watermark Mask Used for Inpainting

C. Pipeline 2: Adaptive Thresholding-Based Approach

This pipeline is suited for softer, embedded watermark patterns, particularly in images with non-uniform illumination or detailed backgrounds.

- 1) The input image is first converted to grayscale.
- 2) Adaptive Gaussian thresholding is applied to segment regions based on local intensity variations.
- 3) A morphological closing operation smoothens and fills gaps in the segmented mask.
- 4) The cleaned mask is used with the inpainting algorithm to remove the watermark.
- 5) A sharpening filter enhances the final output by restoring texture lost during inpainting.

This method excels in cases where the watermark has a low visual footprint and blends with textured backgrounds.

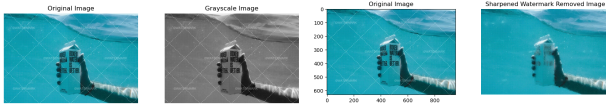


Fig. 4. Left: Original and Grayscale Image. Right: Sharpened Final Output After Inpainting

D. Software Libraries and Tools

This project utilizes several core Python libraries:

- **OpenCV:** Primary library for image reading, preprocessing, edge detection, thresholding, morphology, and inpainting.
- **NumPy:** Used for matrix operations, image transformations, and pixel-wise computations.
- **Matplotlib:** Used to visualize original, intermediate, and final images to evaluate and debug the pipeline stages.

IV. RESULTS

A. Evaluation Approach

To assess the effectiveness of the proposed watermark removal pipelines, two test images containing different types of watermarks were selected. The evaluation primarily focused on a qualitative analysis, where visual comparisons were made between the original, masked, and final inpainted images. The goal was to determine the effectiveness of segmentation, the accuracy of mask generation, and the overall restoration quality after inpainting.

The two pipelines—edge detection-based and adaptive thresholding-based—were compared based on their ability to detect and remove watermarks while preserving image quality. Factors considered in the comparison included the completeness of watermark removal, the presence of artifacts, and the retention of image texture and detail.

B. Dataset

Two real-world images with watermarks of varying complexity and transparency were obtained from online sources. One image contained a structured watermark with clearly defined boundaries, while the other had a semi-transparent, blended pattern watermark over a textured background. The dataset was not predefined or structured, as the study was conducted in an exploratory manner to evaluate the adaptability of each pipeline to diverse watermark types.

C. Qualitative Results

Edge Detection Approach: The edge detection pipeline performed well in cases where the watermark exhibited sharp, well-defined contours. By leveraging Canny edge detection and morphological processing, the method effectively extracted structured watermarks and generated a refined mask for inpainting. However, its performance degraded when dealing with semi-transparent watermarks, as the edges were not distinct enough for proper segmentation. This resulted in incomplete mask generation and residual artifacts in the inpainted images.

Adaptive Thresholding Approach: The adaptive thresholding method demonstrated strong performance in handling semi-transparent watermarks by dynamically adjusting segmentation thresholds across different regions of the image. This approach proved particularly effective in detecting faint, blended pattern watermarks that lacked strong edge features. However, it struggled with complex backgrounds, often introducing unintended artifacts or incorrectly classifying textured regions as part of the watermark. In such cases, inpainting led to some loss of detail, making the restored areas appear slightly blurred.

D. Effectiveness of Approaches

The comparative analysis between the two approaches highlights distinct advantages and limitations. The edge detection-based method excelled in cases where the watermark exhibited clear boundaries, as it accurately extracted contours and facilitated clean inpainting. However, in images where the watermark was embedded within the background, it failed to isolate the watermark completely, resulting in incomplete removal.

Conversely, the adaptive thresholding approach provided superior segmentation of soft, semi-transparent watermarks by adjusting threshold values based on local intensity variations. While this method successfully removed low-opacity watermarks, it occasionally introduced noise in textured areas, leading to noticeable inconsistencies in the inpainted regions. Despite this, the sharpening filter helped enhance details lost during inpainting, partially mitigating the blurring effect.

The results demonstrate that no single method is universally optimal; instead, the choice of pipeline should be based on the nature of the watermark. Structured, high-contrast watermarks are best handled with edge detection, whereas blended pattern, low-contrast watermarks benefit more from adaptive thresholding.

E. Example Outputs

Figure 5 illustrates the results of the watermark removal process using the edge detection approach. The left image shows the original version with a structured watermark, while the right image presents the final output after inpainting. The removal was largely successful, with minor residual artifacts visible in highly textured regions.

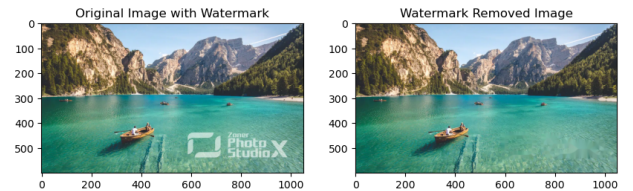


Fig. 5. Original Image with Watermark vs. Watermark Removed Image using Edge Detection

Figure 6 displays the results obtained using the adaptive thresholding pipeline. The semi-transparent watermark was

effectively removed, but some texture inconsistencies are visible in the reconstructed regions. The sharpening filter helped restore some details, making the final output visually cleaner.

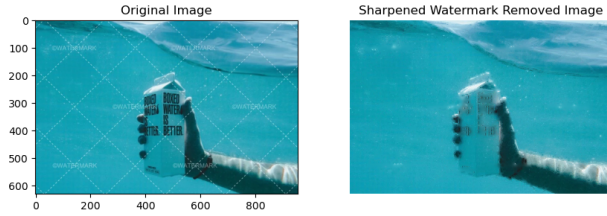


Fig. 6. Original Image with Watermark vs. Sharpened Watermark Removed Image using Adaptive Thresholding

The findings suggest that future improvements could involve combining both techniques in a hybrid model, where edge-based and threshold-based segmentation methods are applied selectively based on watermark characteristics. Additionally, incorporating deep learning-based inpainting models could enhance the texture reconstruction process, reducing artifacts and improving overall image quality.

V. CONCLUSION

This study presents a comparative analysis of edge detection and adaptive thresholding methods for watermark removal. While edge detection proved effective for structured watermarks, adaptive thresholding worked better for soft, blended pattern watermark. Inpainting successfully reconstructed missing areas, but future work can focus on improving texture consistency using deep learning-based approaches such as generative adversarial networks (GANs). Further research can also explore hybrid approaches integrating frequency domain watermark suppression techniques.

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